Lab 2 Worksheet Report

CIFAR-10 CNN Training with Gradient Flow and Weight Update Visualization

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Date: 1 February 2026

Important Links

- **WandB Dashboard:** https://api.wandb.ai/links/shivamkenche-indian-institute-of-technology 9c0kqdrh
- **GitHub Repository:** https://github.com/kingkenche/MLOps-Shivam_Madhav_Kenche-M25CSA028/tree/shivam_m25csa028_lab2_worksheet
- **WandB Project:** cifar10-cnn-training
- **Run Name:** resnet18-baseline

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1 Introduction

This report presents ResNet-18 CNN trained on CIFAR-10 with comprehensive gradient flow and weight update monitoring via WandB. Includes custom dataloader, FLOPs analysis, and 30-epoch training.

2 Model Architecture & Implementation

2.1 ResNet-18 for CIFAR-10

Component	Configuration
Input	3×32×32 RGB images
Initial Conv	3×3, 64 filters, stride=1 (modified for CIFAR-10)
Layers 1-4	2 BasicBlocks each (64→128→256→512 channels)
Output	Global Avg Pool → FC(512, 10)
Parameters	11.17M (11,173,962)
FLOPs	556.66M (556,659,712)

Key Modification: 3×3 initial conv (vs 7×7) and no MaxPooling to preserve spatial resolution for 32×32 inputs.

2.2 Custom DataLoader & Training

Augmentation: RandomCrop(32, pad=4), RandomHorizontalFlip(0.5), ColorJitter(0.2)
Normalization: CIFAR-10 mean/std
Optimizer: SGD (momentum=0.9, weight_decay=5e-4)
Learning Rate: 0.1 with Cosine Annealing
Batch Size: 128 — **Epochs:** 30 — **Hardware:** NVIDIA A30

3 Training Results

Metric	Value
Best Val Accuracy	**93.22%** (Epoch 29)
Final Val Accuracy	93.17%
Final Train Accuracy	99.22%
Training Time	14 minutes (836s)

Progression: Epochs 1-5: 80% → Epochs 6-15: 90% → Epochs 16-30: 92-93%

4 Gradient Flow Analysis

Monitoring: Every 100 steps — **Metrics:** Gradient norm, mean, std per layer

4.1 Key Findings

Layer Type	Gradient Norm Range	Status
Early (conv1, layer1)	0.076-0.102	* Healthy
Middle (layer2, layer3)	0.112-0.241	* Highest (expected)
Deep (layer4, fc)	0.032-0.175	* Residual working

Observations:

- No vanishing gradients (all norms greater than 0.03)
- No exploding gradients (all norms less than 0.3)
- Residual connections ensure that early layers receive gradients
- BatchNorm stabilizes gradient flow

4.2 Visualizations

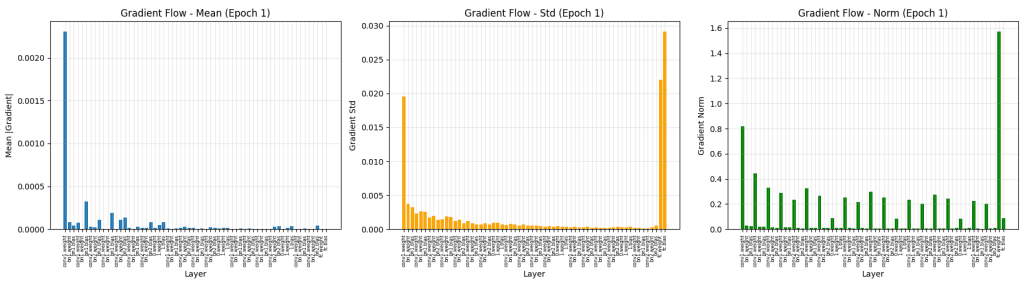

Figure 1: Gradient Flow - Early Training (Step 0)

Progression: Stable gradients from initialization through final epoch; middle layers consistently show higher magnitudes.

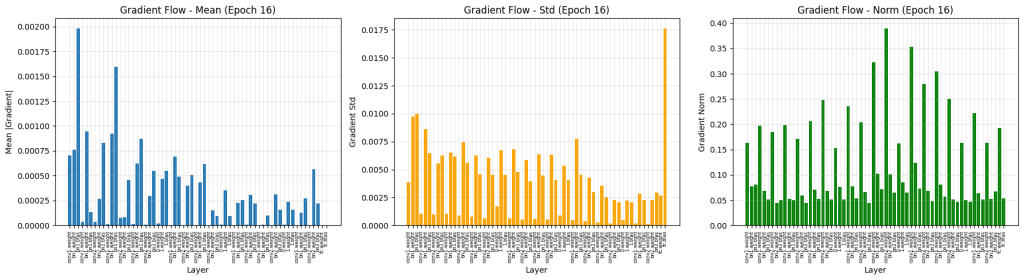

Figure 2: Gradient Flow - Mid Training (Step 6000)

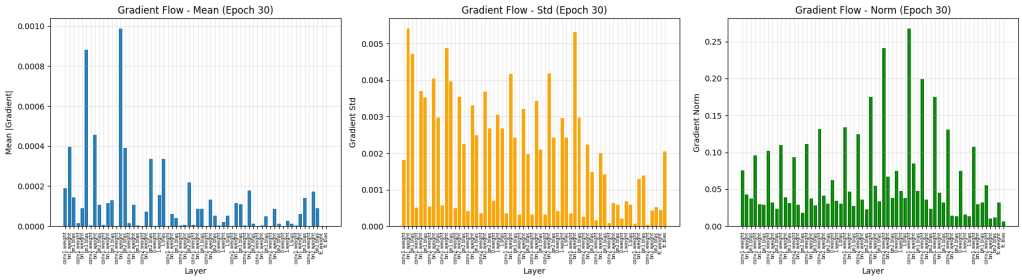

Figure 3: Gradient Flow - Final Training (Step 11599)

5 Weight Update Analysis

Monitoring: Every 100 steps — **Metrics:** Weight norm change, relative change per layer

5.1 Key Findings

Layer	Norm Change	Relative Change
conv1	0.00336	0.00073
layer3.0.conv2	0.01261	0.00117 (max)
fc	0.00124	0.00017 (min)

Observations:

- All layers learning (no frozen weights)
- Cosine annealing: smooth decrease in updates
- Middle layers: largest updates; FC: smallest (well-learned features)
- No oscillations or instability

5.2 Visualizations

Progression: Large initial updates (0.001-0.003) → refined mid-training → fine-grained final (0.0001-0.002).

6 WandB Visualizations

Logged Metrics: Training/validation loss & accuracy, learning rate schedule, gradient flow (per layer, every 100 steps), weight updates (per layer, every 100 steps), model metrics (parameters, FLOPs)

Dashboard: All visualizations available at WandB Link

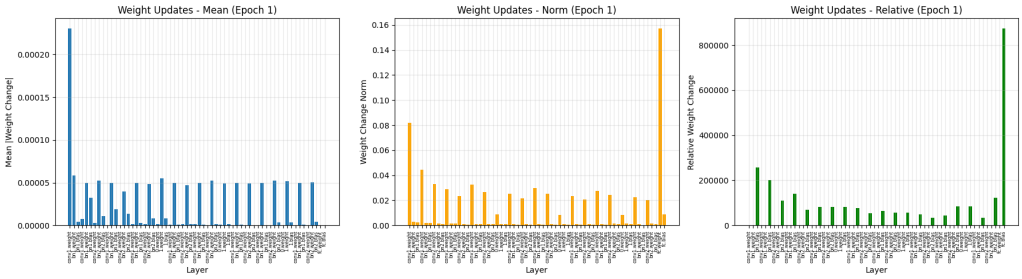

Figure 4: Weight Updates - Early Training (Step 0)

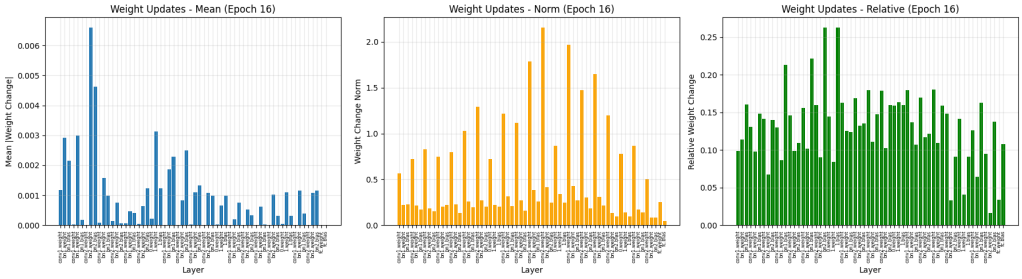

Figure 5: Weight Updates - Mid Training (Step 6000)

7 Conclusions

7.1 Key Achievements

- ResNet-18 adapted for CIFAR-10 (32×32 inputs)
- 93.22% validation accuracy in 30 epochs (14 min)
- Custom dataloader with augmentation
- FLOPs analysis: 556.66M FLOPs, 11.17M parameters
- Comprehensive gradient/weight monitoring via WandB

7.2 Key Insights

Architecture: Skip connections prevent vanishing gradients across all 18 layers.

Training: Cosine annealing + augmentation provide stable convergence. Some overfitting (99.22% train vs 93.17% val) but within acceptable range.

Gradient Health: No vanishing/exploding gradients; BatchNorm + residual connections ensure healthy backpropagation.

Weight Learning: All layers actively learn; middle layers show largest updates, FC layer smallest (feature hierarchy).

7.3 Future Improvements

- Advanced augmentation (AutoAugment, CutOut)
- Learning rate warmup
- Label smoothing
- Extended training

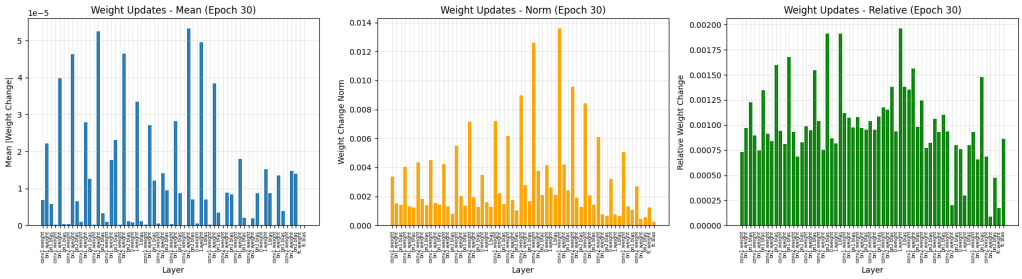

Figure 6: Weight Updates - Final Training (Step 11600)

Appendix: Project Structure

```
Assignment-2/
 src/
    dataloader.py        # Custom CIFAR-10 dataloader
    model.py             # ResNet-18 implementation
    flops_counter.py     # FLOPs counting
    train.py             # Training with WandB
    utils.py             # Gradient/weight trackers
 train_model.py           # Main script
 requirements.txt         # Dependencies
```

Trained models excluded from Git per assignment requirements.

Report generated on 1 February 2026